

Corporations as the State: Urbanisation, Women’s Empowerment, and Long-Run Development in the Copperbelt

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Abstract

Do corporate labour strategies leave persistent legacies long after the firm disappears? I examine two concessions in the Central African Copperbelt: the Union Minière du Haut-Katanga (UMHK) and the Rhokana Corporation (RC). While both firms extracted resources under comparable profit constraints, the UMHK minimized labour turnover by adopting “Labour Stabilisation Programmes” that incentivised permanent family settlement. In contrast, the RC maintained a transient, circular workforce. Using a spatial difference-in-discontinuities design, I show that UMHK’s policy generated urban agglomeration and persistent development gains: individuals inside the former concession exhibit two additional years of education, 30 percentage points higher literacy, and are one standard deviation wealthier. The RC border displays no discontinuity, serving as a falsification test that rules out mineral endowment as the driver of growth. By monetising the family unit and formally integrating women into the labour contract, the UMHK left a distinct gendered legacy: women inside the former concession today report greater reproductive autonomy, reduced tolerance for domestic violence, and increased authority over household finances. These findings illustrate how corporate labour institutions determine long-run development trajectories.

Keywords: Colonial Institutions, Concessions, Corporate Governance, Urbanisation

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1 Introduction

Extractive colonial institutions are widely understood to undermine long-run development (Acemoglu et al., 2001; Dell, 2010). Private concessionary companies, granted state-like powers to exploit vast territories, are typically characterized as the epitome of this predatory model (Lowes and Montero, 2021a). Yet, this paper documents a “paradox of paternalism”: a case where a private corporation, motivated strictly by profit maximisation, generated robust, positive, and persistent development outcomes. I argue that this divergence is driven by a critical, often overlooked nuance in institutional design: the structure of the labour contract. While the standard extractive model treated labour as a transient, disposable input, I show how a corporate strategy incentivising permanent family settlement effectively solved a coordination problem, generating agglomeration economies and human capital accumulation that outlast the firm itself.

I isolate the causal impact of corporate strategy by exploiting a unique natural experiment in the Central African Copperbelt. This region, divided between the Belgian Congo (now DRC) and Northern Rhodesia (now Zambia), hosted two comparable mining giants: the Union Minière du Haut-Katanga (UMHK) and the Rhokana Corporation (RC) (Ferguson, 1999; Van Reybrouck, 2010; Laporte, 2017; Munene, 2018). Operating in a contiguous geographic setting with similar mineral deposits, both corporations faced identical profit constraints and sectoral shocks, most notably the Great Depression and local labour scarcity (Juif and Frankema, 2018). However, their institutional responses to these shared pressures diverged sharply. While the UMHK pivoted to a “Labour Stabilisation” model – encouraging human capital while investing in healthcare and family housing to secure a permanent workforce – the RC entrenched the Southern African model of extractive, circular migration based on short-term seasonal contracts (Parpart, 2018). This stark variation in corporate governance within a homogeneous setting allows me to disentangle the effects of specific labour institutions from the broader impact of colonial regimes, mineral wealth, and the concessionary model as such.

To estimate these legacies, I employ a spatial difference-in-discontinuities design ([Grembi et al., 2016](#)). By benchmarking the discontinuity in outcomes at the UMHK border against the discontinuity at the RC border, this approach isolates the specific impact of corporate policy from the broader institutional differences between the Belgian and British colonial regimes. The validity of this design rests on the exogeneity of the concession boundaries ([Keele and Titiunik, 2015](#)). I document that these borders, originally delineated as vast prospecting zones, do not systematically align with pre-colonial ethnic territories or the location of mineral deposits. Validating this, formal tests indicate balance at both concession boundaries, with continuity across a wide variety of geographic and pre-concession demographic covariates. Furthermore, the overarching colonial border separating the two regions was itself arbitrarily determined by European powers in 1894 ([Gordon, 2001](#)), preceding the discovery of the Copperbelt’s true mineral wealth.

Analysis of georeferenced data from the 2007 and 2014 Demographic and Health Surveys (DHS) reveals a striking divergence in long-run development. Inside the former UMHK concession, outcomes are robustly positive. Households are approximately one standard deviation wealthier, and human capital accumulation is markedly higher: individuals achieve, on average, two additional years of schooling and are 30 percentage points more likely to be literate. Health outcomes mirror this pattern, with women and children ranking around 15 percentiles higher in height-for-age. This anthropometric evidence is particularly telling: it confirms that the concession provided a superior biological standard of living during childhood (See [Steckel, 1995](#); [Fogel, 2004](#)), suggesting that the results are driven by historical institutional quality rather than contemporary selective migration. Consistent with this, alternative specifications that split the sample by current immigration status confirm that development gains are largest among the native population today. Notwithstanding, I cannot rule out intergenerational effects of endogenous sorting as a channel driving development.

In sharp contrast, I find no comparable discontinuity in development outcomes at the former RC boundary. This null result serves as a falsification test: it weakens the hypothesis

that mineral endowments drive the observed prosperity. It suggests that the RC’s extractive model was institutionally indistinguishable from the prevailing colonial framework in Northern Rhodesia; effectively, corporate extraction was the default, not an aberration. This lack of a unique RC legacy reinforces the interpretation that it was the UMHK’s specific deviation into paternalism, rather than the general presence of a corporation, that generated long-run development. Formally, the spatial difference-in-discontinuities analysis confirms that the positive effects at the UMHK border are statistically distinct from the null effects at the RC border, isolating the specific impact of the labour stabilisation model.

I trace this divergence to the firms’ opposing institutional responses to the binding constraint of labour scarcity. After the Great Depression, both corporations faced intense pressure to secure a stable labour supply – due to the copper boom driven by WWII re-armament – yet they adopted diametrically opposing strategies ([Juif and Frankema, 2018](#)). The UMHK pivoted from early coercive practices to a comprehensive Labour Stabilisation model, using long-term contracts and extensive non-wage benefits—including company – provided housing, healthcare, and schooling – to incentivise the permanent resettlement of miners and their families ([Higginson, 1989](#)). In sharp contrast, the RC adopted the South African model of circular migration, favouring short-term seasonal contracts and actively discouraging permanent African settlement to prevent “detribalisation” ([Wilson, 1942](#)). Furthermore, the RC enforced a rigid colour bar that explicitly reserved skilled positions for Europeans and locked African labourers into low-skill roles ([Berger, 1974](#); [Butler, 2007](#)), effectively blocking the local accumulation of human capital.

I identify the primary channel of persistence as a fundamental shift in the region’s spatial equilibrium. Standard economic theory suggests that high labour mobility maximises allocative efficiency. However, my results suggest a potential channel where the UMHK’s strategic introduction of labour rigidity – anchoring workers into specific locations via paternalism – solved a coordination failure and spurred rapid urbanisation. By reducing turnover, the firm may have effectively subsidised the fixed costs of city formation. This generated persis-

tent agglomeration economies and created a path-dependent demand for public goods absent elsewhere in the region. Its legacy remains physically visible today: the former concession area exhibits significantly higher nightlight density ([Henderson et al., 2012](#); [Donaldson and Storeygard, 2016](#)) and built-up surface area, with households nearly 80 percentage points more likely to have access to electricity and piped water. This prosperity persists despite a comparative deterioration in the road network relative to the control group, indicating that agglomeration economies, rather than transport infrastructure, drive the effect. In contrast, the RC’s reliance on circular migration hindered this process of urban formation, leaving behind a fragile spatial structure that failed to sustain development after the mining industry declined ([Ferguson, 1999](#); [Mususa, 2012](#)). This divergence suggests that corporate-driven urbanisation acts as a powerful channel for institutional persistence, transforming temporary corporate incentives into permanent spatial advantages.

The divergence in corporate strategy also created distinct social fabrics by fundamentally shifting the balance of power within the household. Consistent with collective household models ([Chiappori, 1992](#); [Anderson and Eswaran, 2009](#)), the UMHK’s family-settlement model structurally enhanced female intra-household bargaining power: by making high-value non-wage benefits – such as housing, marriage bonuses, and payments for newborns – contingent on the presence of a spouse, the corporation effectively integrated women as stakeholders in the labour contract. In contrast, the RC’s circular migration model entrenched male-dominated transient communities, often characterised by precarious gender relations ([Chauncey, 1981](#); [Parpart, 1986](#)). I show that this historical shock to the household equilibrium persists: women inside the former UMHK concession exhibit significantly higher reproductive autonomy – more than doubling the usage of modern contraception – greater control over household finances, and a 25 percentage point reduction in the justification of domestic violence. These findings suggest that corporate paternalism, while designed to secure labour, inadvertently generated a durable institutional legacy of female empowerment.

This paper contributes to the political economy of colonial institutions by resolving an

ambiguity in the legacy of private rule. While a seminal literature establishes the persistence of colonial institutions ([Acemoglu et al., 2001](#); [Huillery, 2009](#); [Dell, 2010](#); [Wantchekon et al., 2015](#)) and the role of pre-colonial characteristics ([Nunn, 2008](#); [Michalopoulos and Papaioannou, 2013](#); [Lowes and Montero, 2021b](#)), recent work has focused specifically on the distinct role of private colonialism, often finding negative aggregate effects driven by coercion ([Guardado, 2018](#); [Lowes and Montero, 2021a](#); [Chiovelli et al., 2025](#)). I advance this literature by isolating the heterogeneity of corporate governance. By exploiting a setting where two firms shared the same extractive mandate but diverged in their labour strategy, I distinguish the effects of corporate paternalism from the extractive penalty of mining itself. This offers a rare counter-example to the predatory logic of extraction ([Olson, 1993](#); [Acemoglu and Robinson, 2008](#); [Sánchez De La Sierra, 2020](#); [Henn et al., 2025](#)), demonstrating that profit-maximising firms can generate public goods and institutional quality that rival the state under specific labour constraints ([Dell and Olken, 2020](#); [Méndez and Van Patten, 2022](#)). These findings further provide micro-evidence for the view that the quality of institutions mediates the resource curse ([Mehlum et al., 2006](#); [Robinson et al., 2006](#); [Andersen and Aslaksen, 2008](#); [Ross, 2015](#)).

Second, I provide a micro-founded mechanism for the persistence of spatial inequality: path-dependent urbanisation. While the existing literature often attributes local development to transportation infrastructure ([Jedwab and Moradi, 2016](#); [Storeygard, 2016](#); [Donaldson and Hornbeck, 2016](#); [Donaldson, 2018](#); [Banerjee et al., 2020](#); [Jedwab and Storeygard, 2022](#)), I argue that infrastructure without settlement may be insufficient. Consistent with theories of path dependence ([Krugman, 1991](#); [Glaeser et al., 1992](#); [Michaels, 2011](#); [Bleakley and Lin, 2012](#)), I propose that the UMHK’s labour policy solved the coordination problem of city formation, creating a density of human capital that persisted long after the corporate incentives disappeared. This contrasts with the fragility of the RC’s company camps ([Mususa, 2012, 2014](#)), supporting the view that agglomeration economies – not just mineral rents or railway lines – are a potential driver of persistent local development in resource-rich

settings (Melo et al., 2009; Glaeser, 2011; Duranton and Puga, 2014; Gollin et al., 2016; Jedwab et al., 2017; Donovan et al., 2024).

Finally, this paper bridges the gap between resource economics and the cultural origins of gender norms. A prominent literature, exemplified by Ross (2008), argues for a gender resource curse, where extractive industries reduce female labour force participation and political power. While recent empirical work suggests mining can shift local labour markets for women (Kotsadam and Tolonen, 2016; Benschaul-Tolonen, 2019), I identify a distinct channel: corporate paternalism. My findings align with the logic of Qian (2008) and Alesina et al. (2013), which suggest that shifting the relative economic value of women alters their social status. By explicitly monetising the family unit through housing and marriage bonuses, the UMHK model generated a positive shock to female bargaining power. This provides causal evidence that corporate institutional design can inadvertently engineer durable shifts in cultural norms, specifically regarding domestic violence and reproductive autonomy (See Bailey, 2006; Bailey et al., 2012; Tur-Prats, 2019; Bau, 2021; Ojha and Babbar, 2024), challenging the deterministic view of the resource curse for women.

The remainder of the paper proceeds as follows: Section II details the historical context; Section III describes the data; Section IV presents the identification strategy and results; Section V explores the mechanisms of urbanisation and gender; and Section VI concludes.

2 Historical Framework

2.1 Copperbelt Concessionary Companies

The 1884-1885 Berlin Conference established general areas of influence in Africa, while specific borders were later decided during bilateral negotiations. Figure 1 shows the colonial border of the Congo pedicle, resulting from the Anglo-Belgian agreement of 12 May 1894. This division between the BSAC and the CFS initiated fundamental changes in the region due to differences in official colonial policy: the British reliance on Indirect Rule and

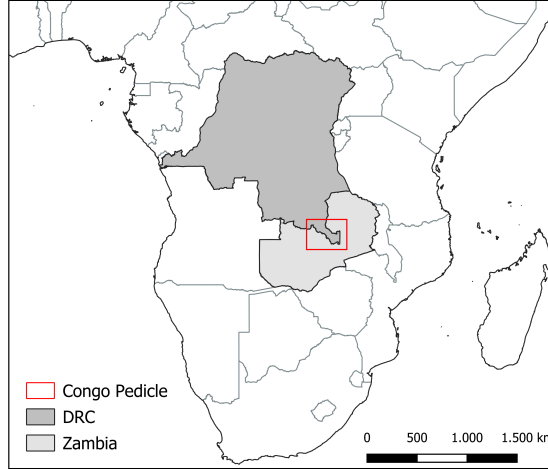


Figure 1: The Congo Pedicle in the Central African Copperbelt

the Belgian paternalistic principle of *dominer pour servir* ¹. These heterogeneous colonial state-corporate relationships influenced the corporations' institutional responses to shared pressures, including fluctuating copper prices and labour scarcity.

In 1891, the province of Katanga was initially set up as a concession under the name *Compagnie du Katanga* to deter British interference in the region. By 1899, King Leopold II had entrusted its administration to the private *Comité Spécial du Katanga* (CSK). Operating largely independently and reporting directly to Brussels, the CSK later partnered with British Tanganyika Concessions Limited (TCL) in 1906 to establish the *Union Minière du Haut Katanga* (UMHK). The UMHK was granted extensive private concessions for copper exploitation in southern Katanga. The corporation retained this copper mining monopoly even after the Belgian state assumed formal control over the Congo in 1908 ².

Katanga's rich surface copper oxide deposits (7 to 15 per cent copper) facilitated easy extraction in the form of low-tech open pit style mines (Coleman, 1971). Production began in 1912 and the UMHK became the 3rd largest copper producer in the world by the 1920s, accounting for 6% of world output (Perrings, 1979). It also became the number one global

¹A phrase coined in 1948 by the Belgian Congo's Governor-General Pierre Ryckmans, justifying authoritarian colonial rule with rapid social and economic development (Ryckmans, 1948).

²The CFS of King Leopold II was taken over by the Belgian government in 1908 under international pressure, leading to the creation of the Belgian Congo.

cobalt producer and an early uranium producer, holding a virtual world monopoly until WWII ³. Figure 2 illustrates the state of all mining concessions at the eve of Congolese independence. Notably, the UMHK concession area remained unchanged throughout its 60 years of operation (1906-1966).

In parallel, Northern Rhodesia’s early years were similarly shaped by private interests. Founded by the colonial entrepreneur Cecil John Rhodes, the BSAC governed the colony until 1924. Northern Rhodesian mineral deposits were first discovered in 1902 at Bwana Mkubwa (now Ndola). Although the BSAC created the Nkana concession in 1911, these were initially low-grade oxide ores unprofitable for large-scale mining (Slinn, 1971). The local copper mining industry took off from the mid 1920s, about 15 years after it first developed in Katanga. This was driven by advancements in froth flotation mineral processing that enabled the profitable extraction of abundant sulphide deposits in Nkana, despite their lower percentage of pure copper compared to high yield Katangan veins (Mouat, 1996; Burdette et al., 1990). At this time, the Nkana concession was being extensively drilled and developed by the Bwana Mkubwa Company, the Rhodesia Congo Border Company (RCBC), and the Anglo-American Company (AAC) (Munene, 2018).

In 1931, these three Copperbelt mining companies consolidated into the Rhokana Corporation (RC) – named after Rhodesia and Nkana. This historical merger was driven by several motives: to more easily raise fresh capital, secure a better bargaining position for output restriction discussions with other large copper producers, and negotiate more effectively with the BSAC and Northern Rhodesian administration (Alford and Harvey, 1980; Munene, 2018). The RC’s attention centered on the Nkana concession, discontinuing the development of the Nchanga mines and any active prospecting in the RCBC. Figure 3 shows the Nkana concession located within the RCBC at RC’s inception (Brooks, 1944). The corporation became the largest Copperbelt mining company, responsible for over half of Northern Rhodesia’s copper output (Munene, 2018), remaining in continuous operation until 1969.

³The uranium for the Manhattan Project originated for over 90% from Katanga (Laporte, 2017).

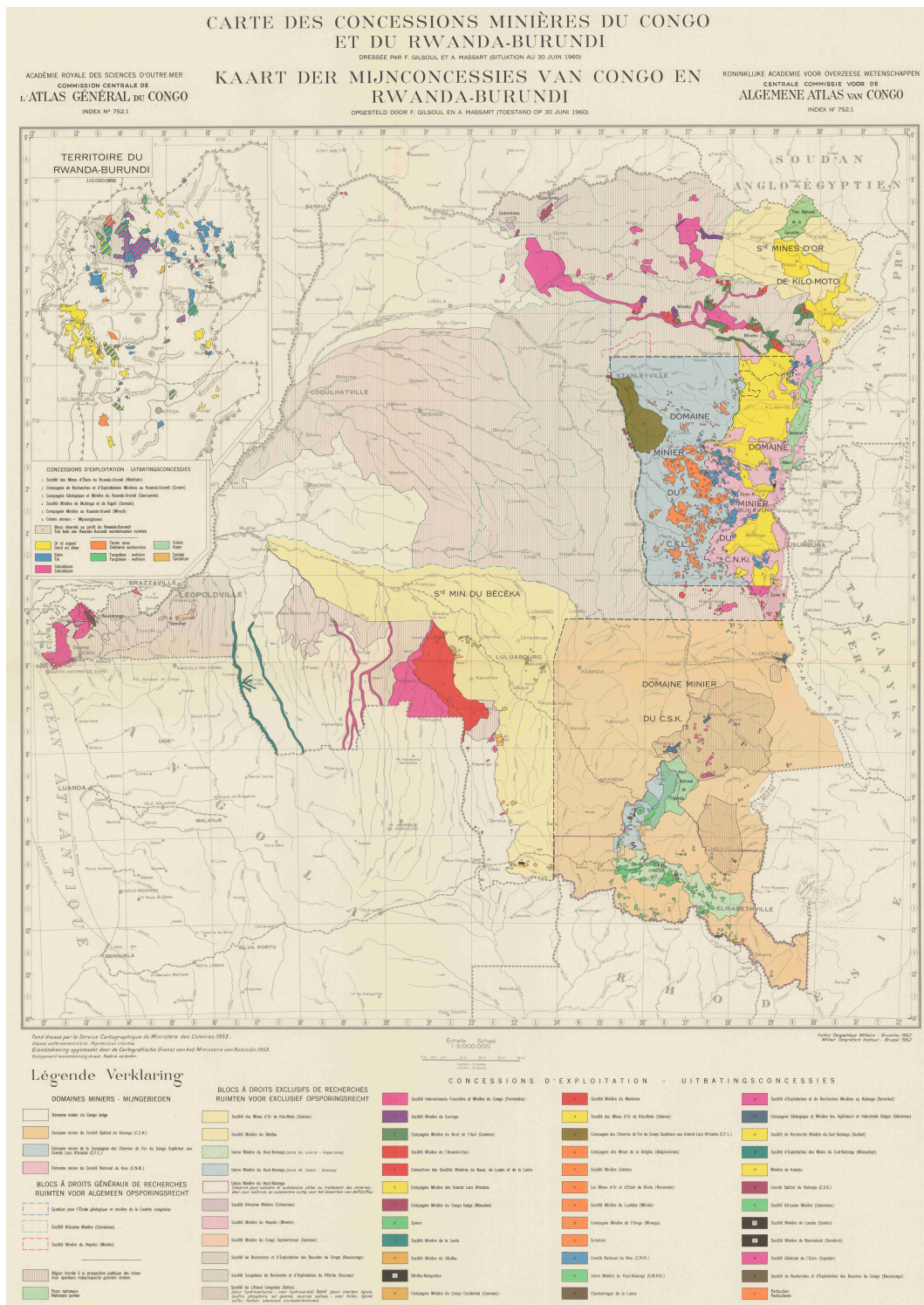


Figure 2: Map of Belgian Congo's Mining Concessions

Notes. The UMHK concession is represented by the dashed green area in the bottom right province of Katanga. The map was created by the Belgian Ministry of Colonies and presents the situation of mining concessions in 1960. This version comes from the Royal Library of Belgium, which was digitised by the Royal Academy of Overseas Sciences.



Figure 3: Map of Northern Rhodesia's Mining Concessions

Notes. The Nkana concession is the small area in the centre north of Northern Rhodesia. The map was created by the Anglo American Company and presents the situation of mining concessions in 1929. This version comes from [Frederiksen \(2013\)](#), from a report by the AAC.

2.2 Colonial Labour Institutions

In October 1929, the New York Stock Exchange crashed, triggering a global economic downturn and financial turmoil. Between 1929 and 1931, global copper prices halved as world consumption declined ([Munene, 2021](#)). This fall in commodity prices, production, and consumption marked a pivotal moment of institutional transformation for the copper industry in both the Belgian Congo and Northern Rhodesia.

Falling profits – driven by rising labour and recruitment costs, coupled with declining metal prices – forced the UMHK to reassess its previously violent, short-term, and high-turnover labour policy ⁴. The labour commission of 1924-25 already recommended es-

⁴Early UMHK operations relied on coerced labour recruited by Robert Williams and Company (TCL subsidiary), which was later managed by the *Bourse du Travail du Katanga* (BTK) – a state-corporate alliance that violently extracted labour from rural areas ([Gordon, 2001](#)).

establishing a more stable, skilled African workforce. In the initial stages of the economic downturn, the mines were able to easily shed their precarious migrant and foreign labour. Further decreased profits during the Great Depression prompted the adoption of a new colonial vision. The UMHK fully embraced its “Labour Stabilisation Programs” – policies aimed at stabilizing its labour supply (Perrings, 1979; Ferguson, 1999). This paternalistic approach permeated nearly every aspect of life within the concession. Miners were expected to sign stable, restrictive, longer-term contracts, binding them to the mines for extended periods. As African workers often left families in their home villages, the UMHK encouraged permanent family resettlement (including wives and children). Family housing was offered as a part of the miners’ benefits. Moreover, women and children were not only accepted to secure necessary labour, but also as a source of both morale and future labour:

“Women and children at U.M. [UMHK] are treated as being the mine’s strength and under their discipline and control. Schools are provided and women participate e.g. in horticulture.”

(Juif and Frankema, 2018)

Investment into children’s education was viewed as both a charitable activity and an investment yielding future returns (the next generation of mine employees). Marriage and childbirth were incentivised: UMHK provided bride prices as signing bonuses for unmarried men and payments for newborns. Juif and Frankema (2018) quantifies labour stabilisation in Katanga: forced labour recruitment fell from 75% of total workers before the programs to only 10% in the 1950s. Females as a percentage of male population in UMHK compounds evolved from 28% in the early 1920s to over 80% near independence. The percentage of married indigenous labourers, the birth rate, and the number of children per 100 workers at the UMHK all doubled over the first decade of labour stabilisation.

Similarly, increased investment in African worker skills was an integral part of the labour stabilisation programs. Such investment was only profitable with sufficiently long-term employment, a three-year threshold was deemed necessary for workers to integrate and justify skill investment. A key motivating factor was that medium skilled black employees were

significantly cheaper than their white counterparts. In 1938, 65% of the workforce was still on contracts shorter than three years. By 1950, this percentage fell to 23.9% (Juif and Frankema, 2018).

In contrast, though the beginnings of large-scale mining in Northern Rhodesia coincided with the Great Depression, threatening to stall the industry in its infancy, its impact was less disruptive for Northern Rhodesia compared to the Belgian Congo (Munene, 2021). Multiple scholars have identified labour as the crucial factor (Perrings, 1979; Butler, 2007; Larmer, 2017). The RC mines relied on a larger share of cheap, abundant, unskilled African labour. Colonial-era scholars contemporaneously highlighted this imbalance, noting its far-reaching effects on African miners' social and economic life (Wilson, 1942; Saffery, 1943).

As the Northern Rhodesian Copperbelt expanded in the late 1920s, the RC quickly adopted labour policies similar to those in South Africa. It featured predominantly seasonal contracts and circular migrant labour, eschewing permanent contracts and resettlement. Workers were encouraged to return to their home villages after their contracts were over. This system was motivated by the Northern Rhodesian government's fears of "detribalisation." Administrators expected that labour stabilisation would accelerate urbanisation, reducing tribal authority and strength. The authorities feared that widespread severance of tribal identity could lead to social unrest.

Beyond contractual organisation, the Copperbelt mines also introduced the South African dual (skilled and unskilled) labour structure, designed to maximise profits (Mhone, 1982). The mining sector required both skilled and unskilled labour in development, construction, and production. Most Africans lacked formal education and could not supply skilled labour, which was reserved for Europeans through a tight industrial "colour bar" (i.e., strong segregation). Unlike their northern neighbours, African workers in Northern Rhodesia were explicitly barred from medium-skilled jobs. Additionally, investment in African labour skills was simply unprofitable due to the use of seasonal, short-duration contracts.

3 Data

To estimate the persistent impact of corporate governance on long-run welfare, I construct a micro-level dataset combining geocoded individual responses from the 2007 and 2014 Demographic and Health Surveys (DHS) for the DRC and Zambia (ICF, 2014). I focus on three dimensions of development: asset wealth, human capital, and biological well-being. Household wealth is measured using the standard DHS asset ownership index and its logarithmic score. Human capital accumulation is captured by individual literacy and total years of schooling. To distinguish local institutional quality from selective migration of healthy adults, I examine anthropometric outcomes – specifically height-for-age percentiles – for women and children. These biological markers serve as a proxy for the disease environment and nutritional standards during the respondent’s early childhood. Table 1 presents summary statistics for all DHS clusters within a 200 km bandwidth of the former UMHK concession boundary.

To isolate the mechanisms of persistence, I construct specific indices of female empowerment and spatial agglomeration. Using the DHS women’s questionnaire, I measure intra-household bargaining power through two variables: Remuneration, a categorical scale ranging from in-kind payments to formal cash wages; and Household Purchases, which captures the respondent’s decision-making authority over major and daily expenditures. I further probe the urban persistence channel using satellite imagery to proxy for economic density. I employ nightlight luminosity data from the VIIRS sensor and built-up surface area data from the Global Human Settlement Layer (GHSL). Complementing these density measures, I include household-level indicators for access to electricity and piped water to track the extent of public utility networks.

I determine treatment assignment by reconstructing the historical geography of the Copperbelt. I manually digitised the concession boundaries using original colonial maps: the *Atlas Général du Congo* (1954) for the UMHK (Académie Royale des Sciences d’Outre-Mer, 1954), and the Rhodesian Anglo American maps (1929) for the RC (Frederiksen, 2013). I

Table 1: UMHK Concession Summary Statistics

Variable	Outside Concession			Inside Concession		
	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
<i>Panel A: Development Outcomes</i>						
Wealth index	5,642	2.531	1.304	5,096	4.354	1.126
Log(wealth score)	5,642	−6.207	8.726	5,096	7.892	8.567
Years of education	5,624	3.031	3.641	5,080	4.878	4.874
Literacy	1,559	0.538	0.499	1,466	0.839	0.368
Woman ht./age pctl.	543	15.117	20.394	479	30.928	27.870
Child ht./age pctl.	565	23.122	34.458	497	33.188	36.157
<i>Panel B: Women's Outcomes</i>						
Modern use (contra.)	1,057	0.017	0.129	982	0.093	0.290
Modern knowledge (contra.)	1,057	0.799	0.401	982	0.883	0.322
Arguing (dom. violence)	1,024	0.663	0.473	957	0.511	0.500
Refusing sex (dom. violence)	989	0.569	0.495	909	0.403	0.491
Remuneration	859	2.073	0.929	548	2.385	0.940
Household purchases	269	0.770	0.719	397	0.776	0.702
<i>Panel C: Urbanisation Outcomes</i>						
Nightlights	5,642	0.036	0.128	5,096	8.214	7.711
Built-up surface	5,642	0.037	0.107	5,096	0.541	0.362
Electricity	5,642	0.041	0.198	5,096	0.595	0.491
Water	5,642	0.122	0.328	5,096	0.684	0.465

Notes. The table reports summary statistics for the sample restricted to observations within 200km of the UMHK concession boundary. *Outside Concession* refers to units with distance < 0 (control group), and *Inside Concession* refers to units with distance > 0 (treatment group). See results tables for detailed variable definitions.

spatially match each DHS cluster to these polygons to define the treatment indicator and running variable (distance to the boundary).

To ensure that the estimated discontinuities reflect institutional differences rather than environmental endowments, I compile a comprehensive vector of geographic and historical covariates. I characterize the physical terrain using digital elevation models (DEM), slope, and ruggedness (Nunn and Puga, 2012). Agricultural potential is controlled for using soil quality data from the Atlas of the Biosphere (University of Wisconsin–Madison), computed from soil carbon and nitrogen levels following Michalopoulos (2012), as well as long-run rainfall volatility from the Global Climate Database. Finally, to account for pre-colonial

institutional differences, I link observations to ethnic characteristics from Murdock’s Ethnographic Atlas (Murdock, 1967). and slave trade exposure data from Nunn and Wantchekon (2011).

Finally, I reconstruct the historical supply of public goods to check for colonial-era investments. For the Belgian Congo, I digitise and georeference historical locations of healthcare facilities and missionary posts from the *Atlas du Congo* using QGIS. For Northern Rhodesia, I utilize the recent dataset by Chiseni (2024), which offers superior spatial precision compared to older missionary compendia (See Jedwab and Storeygard, 2022). I further digitise post-independence transport networks (circa 1968) from the *Perry-Castaneda Map Collection* and modern Congolese infrastructure (2010) from *Le Référentiel Géographique Commun* to track the evolution of transportation infrastructure.

4 Development Outcomes

4.1 Empirical Strategy

I estimate the impact of exposure to colonial concessions and their private corporations on development outcomes using a geographic regression discontinuity (RD) design.

The baseline specification is:

$$Y_{i,c} = \alpha + \beta \text{Concession}_{i,c} + f(\text{geolocation}_c) + \gamma \mathbf{X}_i + \epsilon_{i,c} \quad (1)$$

where $Y_{i,c}$ is the outcome variable for individual i from DHS cluster c . $\text{Concession}_{i,c}$ is an indicator equal to 1 if DHS cluster c is inside the concession area and 0 otherwise. $f(\text{geolocation}_c)$ is the RD polynomial, controlling for smooth functions of geographic location. $\mathbf{X}_i$ is a vector of individual-level covariates, including sex, age, age squared, and survey year. Finally, $\epsilon_{i,c}$ is the error term, with standard errors clustered at the DHS cluster level.

The baseline specification uses a local linear polynomial in distance to the boundary,

estimated separately on each side (Gelman and Imbens, 2019), and follows best practices for spatial regression discontinuities (Calonico et al., 2014a,b; Cattaneo et al., 2016, 2019). This involves calculating the optimal bandwidth through a Mean Squared Error (MSE)-minimizing procedure. For robustness, I also present results using a wider, fixed bandwidth of 200 km, ensuring approximately balanced clusters on either side of the boundary.

The coefficient of interest, β , captures the effect of being located just inside a former colonial concession. Causal interpretation relies on the assumption that concession boundaries are orthogonal to pre-concession local geography and demography. This central continuity assumption is critical for causal identification, as pre-existing differences in geography, culture, or political organisation could otherwise confound contemporary development outcomes (e.g., Nunn, 2008; Michalopoulos and Papaioannou, 2014; Alsan, 2015; Lowes et al., 2017). Under this assumption, any local discontinuities in outcomes can be attributed to colonial corporate exposure.

To assess whether units on either side of the boundary are valid counterfactuals, I show that geographic and precolonial demographic variables vary smoothly around the concession borders. Panel A in Table 2 presents summary statistics and RD estimates for geographic variables near the UMHK concession boundaries. This balance test utilises a 20 km by 20 km cell grid, with variable values averaged for each cell. The RD estimates are derived from a version of specification (1), adapted to the grid-cell level by averaging outcomes within each cell. Standard errors are clustered at the territory level for geographic characteristics.

I find that geographic characteristics vary smoothly around the concession boundaries, supporting the validity of the identification assumption. Moreover, most of these geographic characteristics are also balanced over the entire bandwidths. These results strongly suggest that the concession areas were not significantly different from the surrounding regions in terms of geography prior to the concessions.

For pre-colonial demographics, I use ethnic group data from the Ethnographic Atlas (Murdock, 1967). I also incorporate variables related to the Atlantic and Indian slave trade,

Table 2: UMHK Concession Balance Test

	Control	RD Specification			
	Mean (SD) (1)	Estimate (SE) (2)	Opt. BW (km) (3)	Effective Obs. (4)	Clusters (5)
<i>Panel A: Geography</i>					
Elevation (m)	1,109.67 (253.66)	−2.13 (137.86)	129.18	591	25
Slope	2,245.94 (1,540.81)	−153.03 (702.79)	105.87	520	25
Ruggedness	84.84 (58.69)	−6.19 (26.71)	105.08	519	25
Soil suitability index	96.35 (1.03)	0.05 (0.77)	131.09	606	25
Rainfall (avg., mm)	1,190.33 (86.37)	−10.14 (30.11)	119.72	565	25
Rainfall (std. dev.)	11.80 (10.07)	−0.45 (3.05)	118.77	562	25
Malaria suitability index	7.31 (3.50)	−0.03 (1.91)	136.11	619	25
Navigable river	0.02 (0.15)	0.06 (0.27)	91.10	479	22
<i>Panel B: Demography</i>					
Population density (log)	−0.25 (0.65)	−0.04 (0.32)	108.45	137	5
Total enslaved (exports)	551.16 (913.49)	368.18 (880.60)	180.56	767	17
State centralization	0.85 (0.36)	−0.20 (0.25)	370.13	1115	14
Polygynous	0.77 (0.42)	−0.09 (0.33)	177.98	717	15
Hereditary headman	0.99 (0.10)	−0.00 (0.02)	168.53	616	15

Notes. The table tests for balance in predetermined characteristics at the concession boundary. Panel A reports geographic variables; Panel B reports pre-colonial institutional and demographic characteristics from the *Ethnographic Atlas* (Murdock, 1967). The unit of observation is a 20 km × 20 km grid cell. Column (1) reports the mean (and standard deviation) of the control group within the optimal bandwidth. Column (2) reports the *Conventional* RD coefficient and the *Robust* standard error (in parentheses). Panel A clusters standard errors at the territory level; Panel B clusters at the ethnic group level. Estimates use a local linear regression with a triangular kernel; standard errors and significance levels are robust to the bias-correction procedure. Column (3) reports the optimal bandwidth selected by the MSE-minimizing procedure (Cattaneo et al., 2019). Columns (4) and (5) report the effective number of observations and clusters within this bandwidth.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

created by [Nunn and Wantchekon \(2011\)](#) and aggregated at the ethnic group level. Panel B in Table 2 shows no significant difference in the makeup of pre-colonial demography at or near the UMHK concession boundary. Crucially, the concession boundary does not align with any pre-colonial ethnic group boundaries, further supporting the identification strategy.

A potential concern in this empirical setting is that concession boundaries may correlate with mineral deposits rather than being randomly drawn. However, this does not undermine identification. Concessions were initially granted as prospecting and exploration zones (e.g., *blocs à droits exclusifs de recherches* in the Belgian Congo; prospecting concessions in Northern Rhodesia), and only later expanded to include mine-specific exploitation rights. As a result, concession boundaries do not systematically align with the location of major mineral deposits, reinforcing the validity of the RD design.

4.2 Results

Long-Run Development. Exposure to the UMHK is associated with consistently positive long-term development outcomes across household wealth (Panel A), individual education (Panel B), and health indicators (Panel C), as reported in Table 3. Households within the former UMHK concession area are approximately one standard deviation wealthier, individuals attain on average two additional years of schooling, and literacy rates are substantially higher, by approximately 30 percentage points. Women and children are also significantly taller for their age, consistent with improved early-life health and nutritional conditions. These results indicate persistent effects of past colonial corporate policies on contemporary development outcomes.

This wide range of benefits challenges common narratives about the universally extractive nature of colonial concessions in Africa. The fact that these effects remain observable decades after the collapse of the UMHK underscores the durability of corporate institutional environments. The magnitude of these effects – particularly the sharp increase in education and literacy – suggests not only greater investment but also a shift in household-level human

capital accumulation, likely tied to the UMHK’s efforts to stabilise and incentivise long-term family settlement ([Foster and Rosenzweig, 1995](#)).

Figure 4 presents RD plots illustrating discontinuities at the UMHK boundary for each outcome in Table 3. The running variable is the distance to the UMHK concession boundary. A linear fitted trend is estimated on both sides of the discontinuity at the border (0 km). Each plot shows a sharp discontinuity at the boundary, visually supporting a causal interpretation of the main results.

Women’s Empowerment. The legacy of the UMHK’s paternalistic policies also extends to women’s contemporary outcomes, an uncommon feature in studies of historical institutional legacies. Gender-related outcomes are markedly better within the former UMHK concession. Women’s ability to access and use modern contraception is a key determinant of their broader life outcomes ([Bailey, 2006](#); [Bailey et al., 2012](#); [Bailey and Lindo, 2017](#)). Panel A of Table 4 shows a significantly higher rate of modern contraceptive use in the former concession area. This difference is not driven by greater awareness, as knowledge of modern methods does not vary significantly across the boundary. Importantly, higher contraceptive use is not accompanied by greater acceptance of domestic violence, suggesting broader social empowerment (cf. [Ojha and Babbar, 2024](#)). Panel B of Table 4 presents results on female attitudes towards domestic violence. In general, women residing in the former UMHK concession area find domestic violence less justified. Specifically, they more often believe that refusing sex or arguing with their husband does not justify violence.

Women’s economic outcomes reinforce this positive legacy. Panel C in Table 4 presents gender outcomes in terms of income and expenses. The remuneration variable, serving as an indicator of women’s economic compensation and the formalisation of their work, represents an earnings scale ranging from no payment whatsoever, to payments in-kind, to mixed in-kind and cash payments, to full wages in cash. The variables on the expense side of household finance represent the degree to which women have a final say in the decision-making process

Table 3: UMHK Concession Development Outcomes

	Optimal Bandwidth (1)	Wide Bandwidth (2)	Optimal Bandwidth (3)	Wide Bandwidth (4)
<i>Panel A: Asset Wealth</i>				
	Wealth index		Log(wealth score)	
UMHK	2.054*** (0.789)	1.987*** (0.698)	17.099*** (6.189)	16.628*** (5.642)
Observations	8,992	10,738	8,992	10,738
Clusters	55	61	54	61
Bandwidth (km)	121.59	200.00	122.33	200.00
Mean dep. var.	3.609	3.396	1.920	0.484
Std. dev. dep. var.	1.468	1.525	11.050	11.154
<i>Panel B: Education</i>				
	Years of education		Literacy	
UMHK	2.305** (0.913)	2.328*** (0.821)	0.337** (0.140)	0.358*** (0.126)
Observations	8,962	10,704	2,372	3,025
Clusters	55	61	55	61
Bandwidth (km)	118.84	200.00	111.49	200.00
Mean dep. var.	4.095	3.908	0.717	0.684
Std. dev. dep. var.	4.452	4.369	0.450	0.465
<i>Panel C: Health</i>				
	Woman ht./age pctl.		Child ht./age pctl.	
UMHK	15.769** (6.865)	16.412** (6.612)	17.687* (9.766)	16.138* (9.410)
Observations	808	1,022	867	1,062
Clusters	52	61	53	61
Bandwidth (km)	110.29	200.00	117.71	200.00
Mean dep. var.	24.624	22.528	28.388	27.833
Std. dev. dep. var.	26.117	25.431	35.607	35.603

Notes. The table reports regression discontinuity estimates using a local linear specification with a triangular kernel, estimated separately on each side of the concession boundary. Standard errors are clustered at the DHS cluster level and reported in parentheses. Optimal bandwidths (in km) are selected using the MSE-minimizing procedure from [Cattaneo et al. \(2019\)](#). All regressions control for sex, age, age squared, and survey year. *Wealth score* is the DHS principal component index of asset ownership. *Wealth index* is a categorical variable (1–5) representing quintiles of the wealth score. *Literacy* is a binary indicator for reading ability. *Woman* and *Child ht./age pctl.* represent the respondent’s height-for-age percentile within the entire sample, calculated for the subsets of eligible women and children, respectively.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

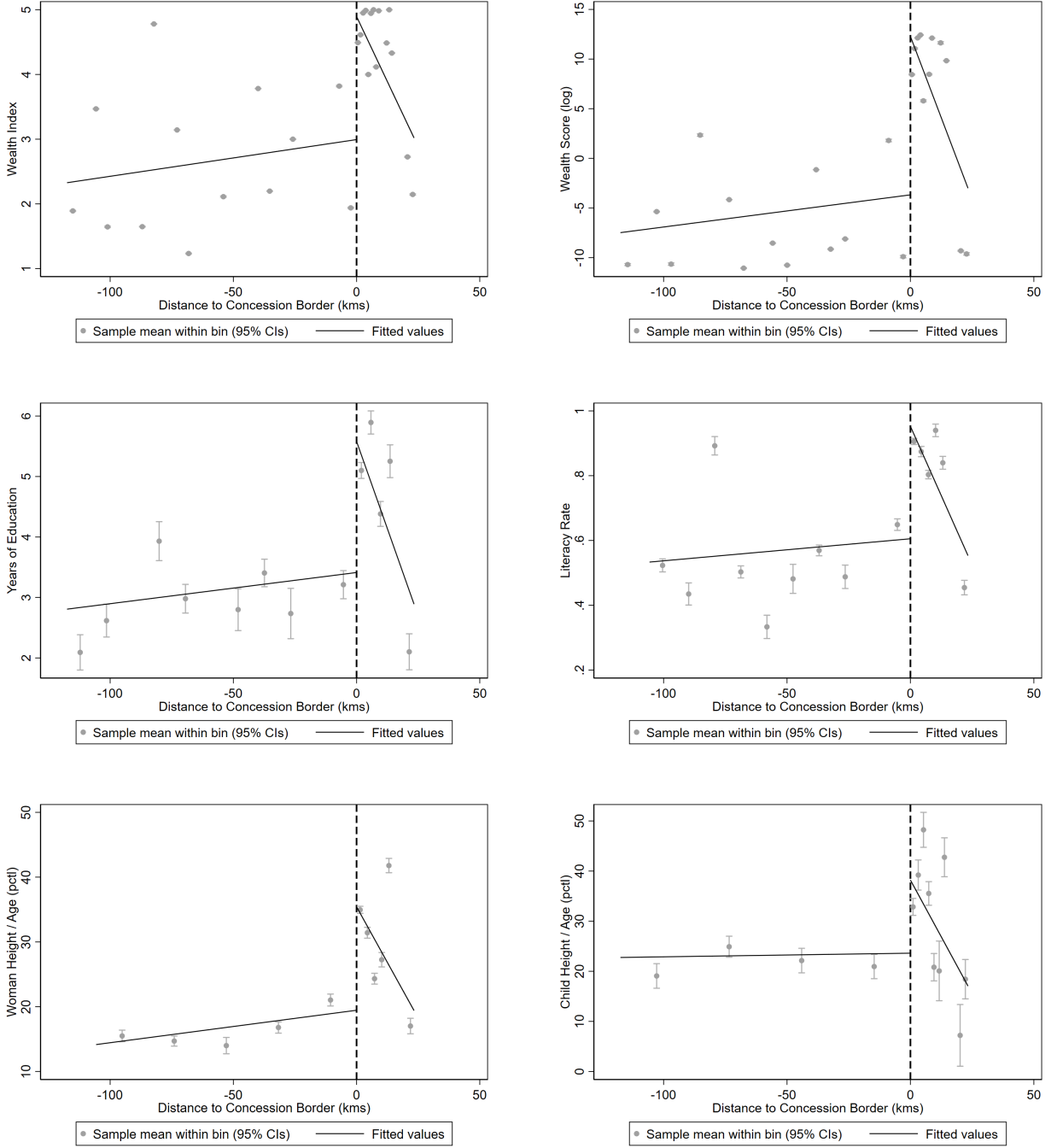


Figure 4: UMHK RD Plots for Development Outcomes

Notes. The figure displays the RD plots for development outcomes and the mean value of each outcome variable at the optimal amount of bins as recommended by Cattaneo et al. (2019) along the running variable (distance to the UMHK concession boundary), including a local linear trend estimated separately on each side of the discontinuity.

of both large household purchases and smaller daily purchases. Women are more frequently compensated for their work, with a higher likelihood of receiving cash payments. They also report greater influence over both major and routine household spending decisions.

Taken together, the evidence suggests that women in the former UMHK area enjoy greater reproductive autonomy, better earnings, and more household decision-making power than women in comparable nearby regions. These gendered effects provide suggestive evidence that paternalistic labour institutions, when tied to family settlement and social provisioning, may have unintended but persistent effects on women’s empowerment.

4.3 Robustness and Sensitivity Checks

Selective Migration. A primary concern with spatial persistence studies is selective migration: do superior outcomes reflect the institutional environment, or simply the sorting of high-human-capital individuals into the former concession area? Accurately determining the magnitude of historical endogenous sorting in the Katanga province and beyond is impossible. However, more recent selective migration is both a larger threat to identification – as it does not constitute a long-run channel itself – and quantitatively identifiable. To address this, I estimate the RD specification separately for ‘natives’ (individuals who report always living in the current location) and migrants.

Table 5 provides results that reject the contemporary endogenous sorting hypothesis. Across all specifications, the discontinuity remains large and statistically significant for the native population. The increased magnitude of the coefficients is driven by data availability, as the question was only answered by women in 2007 (hence no child ht./age pctl.). Notwithstanding, what matters for this robustness check is the relative effect size of these results. Notably, the treatment effects for natives are consistently larger than for migrants. For example, natives within the UMHK concession exhibit an additional 1.5-year increase in education compared to the increase for migrants, and a wealth effect nearly double that of the migrant population.

Table 4: UMHK Concession Women’s Outcomes

	Optimal Bandwidth (1)	Wide Bandwidth (2)	Optimal Bandwidth (3)	Wide Bandwidth (4)
<i>Panel A: Contraception</i>				
	Modern use		Modern knowledge	
UMHK	0.072** (0.034)	0.075** (0.038)	0.145 (0.127)	0.099 (0.081)
Observations	1,849	2,039	1,683	2,039
Clusters	55	61	55	61
Bandwidth (km)	157.81	200.00	120.64	200.00
Mean dep. var.	0.059	0.053	0.858	0.840
Std. dev. dep. var.	0.236	0.225	0.349	0.367
<i>Panel B: Domestic Violence</i>				
	Arguing		Refusing sex	
UMHK	-0.172** (0.081)	-0.177** (0.078)	-0.244*** (0.095)	-0.268*** (0.102)
Observations	1,636	1,981	1,560	1,898
Clusters	53	61	55	61
Bandwidth (km)	122.11	200.00	121.91	200.00
Mean dep. var.	0.461	0.465	0.475	0.489
Std. dev. dep. var.	0.358	0.361	0.500	0.500
<i>Panel C: Income & Expenses</i>				
	Remuneration		Household purchases	
UMHK	1.016* (0.531)	0.872** (0.394)	0.182** (0.074)	0.177** (0.076)
Observations	1,024	1,407	569	666
Clusters	55	61	21	21
Bandwidth (km)	111.72	200.00	114.38	200.00
Mean dep. var.	2.243	2.195	0.766	0.773
Std. dev. dep. var.	0.951	0.945	0.700	0.708

Notes. The table reports regression discontinuity estimates using a local linear specification with a triangular kernel, estimated separately on each side of the concession boundary. Standard errors are clustered at the DHS cluster level and reported in parentheses. Optimal bandwidths (in km) are selected using the MSE-minimizing procedure from [Cattaneo et al. \(2019\)](#). All regressions control for age, age squared, and survey year. *Modern use* and *knowledge* are binary indicators for the usage and awareness of modern contraception methods. *Arguing* and *Refusing sex* are indices for justifying domestic violence based on DHS questions. *Remuneration* is a categorical variable (1–4) ranging from unremunerated to fully cash-based work. *Household purchases* is an index (1–4) measuring the woman’s say in daily and large purchases.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 5: UMHK Concession Development Outcomes by Migration Status

	Native (Optimal BW) (1)	Immigrant (Optimal BW) (2)
<i>Panel A: Asset Wealth</i>		
Wealth index	3.933*** (0.209)	2.266*** (0.801)
Log(wealth score)	25.472*** (0.971)	22.851*** (2.431)
<i>Panel B: Education</i>		
Years of education	7.771*** (0.663)	6.274*** (1.021)
Literacy	0.755*** (0.126)	0.651*** (0.108)
<i>Panel C: Health</i>		
Woman ht./age pctl.	29.672*** (4.932)	20.286*** (7.297)
Bandwidth (Avg, km)	110.8	107.5
Observations (Max)	7,715	5,447
Clusters	21	20

Notes. The table reports regression discontinuity estimates utilizing the Optimal Bandwidth selection procedure (MSE-minimizing) from [Cattaneo et al. \(2019\)](#). *Native* refers to respondents who reported always living in their current location; *Immigrant* refers to those who moved to their current location. All regressions control for sex, age, age squared, and survey year. Standard errors are clustered at the DHS cluster level and reported in parentheses. Bandwidths vary slightly by outcome; the average is reported. Observation counts vary by outcome; the maximum is reported.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Most importantly, the anthropometric results suggest an environmental channel. Native women, whose height is determined by early-life conditions within the former concession, rank nearly 30 percentiles higher in height-for-age than those just outside the border. This confirms that the concession provided a superior biological standard of living during childhood, rather than merely attracting taller adults.

Measurement Error. A different potential limitation of this analysis is that the DHS data clusters are randomly displaced to maintain the confidentiality of the respondents. This random displacement can be up to 2 km for all urban clusters, up to 5 km for 99% of the rural clusters, and up to 10 km for 1% of the rural clusters. This displacement introduces a specific form of non-classical measurement error in the forcing variable (distance to the boundary). While this complicates the standard interpretation, the resulting misclassification of units near the cut-off is, in this context, still most likely to attenuate the estimated discontinuity. Nonetheless, as noted by [Lowes and Montero \(2021a\)](#), the results of an RD approach may be driven by incorrectly assigned clusters right along the boundary.

Given the narrow shape of the UMHK concession area, “donuthole” estimations excluding clusters near the boundary are impractical. However, as the DHS GPS displacement algorithm does not allow clusters to cross internal administrative boundaries, major urban clusters like Lubumbashi, Likasi, and Kolwezi remain correctly assigned within the concession area. This reduces the risk that misclassification drives the observed discontinuities. As this misclassification of treatment status near the boundary is non-classical, it could theoretically bias the results in either direction; however, in spatial settings, this fuzziness at the cut-off most often leads to an attenuation of the true effect. Accordingly, the estimated discontinuities may represent a lower bound on the true impact.

Finally, I verify that the results are robust to using a local quadratic polynomial and alternative bandwidth selection procedures. The main findings appear to be robust to a battery of alternative specifications, shown in Table [A.5](#) in Appendix A.5.

In sum, these findings collectively provide robust evidence of the enduring positive impacts of the UMHK’s presence, not only on overall economic development and human capital but also on various dimensions of women’s empowerment, highlighting the long-term legacy of colonial corporate governance.

5 Mechanisms of Divergence

5.1 Comparative Development Effects

Assessing whether the lasting effects of the UMHK concession reflect its broader corporate strategies requires a comparison to outcomes under the RC concession. I first compare long-run development outcomes across the UMHK and RC concessions, and then examine how differences in corporate governance help explain the divergence.

The relative effect of exposure to each concession is estimated using a spatial difference-in-discontinuities (diff-in-disc) design. This design estimates δ by pooling observations from both regions and fitting linear regressions on either side of each boundary (2×2), following [Grembi et al. \(2016\)](#). The empirical estimation compares the discontinuities at the UMHK boundary to those at the RC boundary, while controlling for differences across the interior and exterior of each concession. The stacked specification is defined as follows:

$$Y_{i,c,b} = \alpha + \beta Concession_{i,c,b} + f_0(geolocation_{c,b}) + \gamma \mathbf{X}_{i,b} + UMHK_b[\zeta + \delta Concession_{i,c,b} + f_1(geolocation_{c,b}) + \eta \mathbf{X}_{i,b}] + \epsilon_{i,c,b} \quad (2)$$

where the coefficient of interest is δ , which captures the differential discontinuity effect of the UMHK boundary relative to the RC boundary. $Y_{i,c,b}$ is the outcome for individual i in cluster c near boundary b . $UMHK_b$ is an indicator equal to 1 if the observation is located near the UMHK concession boundary and 0 otherwise. Other variables follow the baseline specification, with the addition of the concession boundary indicator b , analogous to the post-treatment indicator in standard diff-in-disc settings ([Grembi et al., 2016](#)).

Appendix A.2 shows that geographic and precolonial characteristics vary smoothly across both concession boundaries, supporting the identification strategy. Appendix A.3 also addresses potential contamination from the obsolete Rhodesian Congo Border Concession (RCBC). For the comparison of discontinuities to be valid, the general location of the colonial border must not have been determined by mineral endowments. Otherwise, any observed divergence could reflect initial resource inequality rather than institutional differences.

Historical accounts confirm that the border reflected factors like hunting grounds and river basins, not mineral deposits ([Gordon, 2001](#)). For example, negotiations with local chiefs emphasised hunting rather than control over resources. The Congo pedicle – the only area of real contention – lacked significant mineral deposits. Instead, most copper ore deposits lie further west, where the border was easily determined by the Congo-Zambezi river watershed. Moreover, the region’s true mining potential was only discovered more than a decade after the boundary was drawn.

Panel A of Table 6 reports the diff-in-disc estimates for key household and individual outcomes. Discontinuities are consistently larger at the UMHK boundary, indicating stronger long-run effects. The only exception is children’s height, where the relative difference is less pronounced. This likely reflects the presence of offsetting urban dis-economies, such as congestion and higher disease transmission, further explored in Section 5.4.

The broader pattern supports the interpretation that UMHK’s institutional model generated more durable development gains, while the RC’s approach left fewer lasting benefits for its communities. Crucially, as shown in the full regression discontinuity results in Appendix A.4, the RC boundary exhibits no significant discontinuity in development outcomes. This implies that the observed divergence is driven by the positive legacy of UMHK’s paternalism rather than a negative legacy of the RC. Together, these findings point to differences in corporate institutions – not geography or resources – as the key driver of divergent long-run development trajectories. Moreover, while earlier I controlled for in-migration, differential out-migration from the control group remained a potential source of bias. However, if the

Table 6: Difference-in-Discontinuities Development and Urbanisation Outcomes

	Bandwidth 50 km DiDisc Est. (SE) (1)	Bandwidth 100 km DiDisc Est. (SE) (2)
<i>Panel A: Development Outcomes</i>		
Wealth index	2.273*** (0.061)	1.788*** (0.047)
Observations	18,134	22,696
Log(wealth score)	17.370*** (0.388)	14.231*** (0.288)
Observations	18,134	22,696
Years of education	2.169*** (0.220)	1.870*** (0.179)
Observations	18,064	22,598
Literacy	0.319*** (0.058)	0.285*** (0.045)
Observations	6,511	7,874
Woman ht./age pctl.	15.672*** (5.556)	15.364*** (4.551)
Observations	3,063	3,741
Child ht./age pctl.	15.848** (7.438)	15.448** (6.104)
Observations	2,234	2,908
<i>Panel B: Urbanisation Outcomes</i>		
Nightlights	7.975*** (0.041)	7.391*** (0.032)
Observations	18,134	22,696
Built surface	0.760*** (0.003)	0.601*** (0.002)
Observations	18,134	22,696
Electricity	0.770*** (0.015)	0.692*** (0.016)
Observations	18,112	22,656
Water	0.795*** (0.030)	0.607*** (0.028)
Observations	9,073	11,292

Notes. The table reports Difference-in-Discontinuities (DiDisc) estimates using the RC concession boundary as the baseline counterfactual for the UMHK concession. Columns (1) and (2) report the DiDisc coefficient and standard error (in parentheses) for bandwidths of 50 km and 100 km, respectively. All regressions control for sex, age, age squared, and survey year. *Wealth index*, *Education*, *Literacy*, and *Height/Age Percentiles* follow previous definitions. *Nightlights* (VIIRS) and *Built surface* (GHSL) measure urbanisation intensity. *Electricity* and *Water* are binary indicators for household access.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

RC concession (the null result case) had similar out-migration pressures without the corresponding development boost, it would suggest that out-migration alone cannot explain the magnitude of the UMHK premium. The remainder of the paper therefore focuses on identifying the specific institutional and spatial channels that drove this persistent divergence.

5.2 Corporate Paternalism and Investment

The central institutional divergence between the UMHK and RC concessions lies in their distinct corporate responses to a shared constraint: the inability to use wages as a competitive instrument. These divergent strategies shaped not only immediate labour conditions but also produced lasting differences in urban formation and infrastructural development. While existing literature identifies labour policy as a key determinant of colonial institutions' developmental legacies ([Chiovelli et al., 2025](#)), this paper highlights a novel, indirect channel: corporate labour strategies influenced the pace and permanence of urbanisation, thereby shaping the spatial distribution and durability of public infrastructure.

Wage Coordination. The most direct channel through which firms could have differentiated their labour policies was through wage setting. However, a system of wage coordination was implemented across the Central African Copperbelt in the early 1930s. Mining corporations in both the Belgian Congo and Northern Rhodesia agreed to harmonise pay scales to avoid competition for labour, rendering wages locally non-instrumental as a recruitment tool. Importantly, this coordination did not suppress overall compensation: Copperbelt miners remained among the best-paid manual labourers in Sub-Saharan Africa ([Juif and Frankema, 2018](#)).

Paradoxically, this suppression of price competition (wages) became the catalyst for institutional deepening. Restricted from offering higher pay, the UMHK was compelled to compete on 'quality of life,' effectively transforming the corporation into a provider of public goods to secure its labour supply. The Belgian colonial vision thus emphasised paternalism,

while the British approach was more narrowly extractive. As a result, non-wage benefits began to diverge markedly from the late 1930s.

Healthcare. The UMHK invested heavily in healthcare provision. Column (1) of Table 7 indicates a higher density of healthcare facilities and apothecaries within the former UMHK concession area relative to the surrounding region in 1953. By contrast, similar incentives were largely absent in the RC concession, where short-term labour contracts undermined the rationale for long-term welfare investment. Consequently, the RC provided only limited healthcare services, reflected in the lower concentration of facilities.

Education. Documenting direct corporate investment in education is more challenging, as these services were primarily outsourced to missionaries. Columns (3-4) of Table 7 offer only weak and statistically insignificant evidence of systematic differences in missionary presence. This suggests that missionary presence may be an incomplete proxy for educational investment, or that the large, persistent gains in education were likely driven by a demand-side shift rather than purely supply-side religious infrastructure. The UMHK’s “Labour Stabilisation Programs,” by fostering permanent family resettlement, likely created a stable population with a higher incentive to invest in human capital, a channel consistent with the broader findings on community stability.

5.3 Gender and Household Structure

In addition to direct impacts on human capital, differences in labour policy influenced broader demographic patterns. Corporate labour strategies fundamentally shaped migration decisions, encouraging not only individual labour migration but also the migration of miners’ families into the UMHK concession. This early divergence in settlement policy generated a persistent shift in intra-household bargaining power.

Table 7: UMHK & RC Concessions Labour Policy

	Healthcare		Missionary Posts	
	Facilities (1)	Apothecaries (2)	in 1924 (3)	in 1950 (4)
<i>Panel A: UMHK Concession</i>				
UMHK	0.033** (0.013)	0.102** (0.039)	0.053 (0.037)	0.074* (0.040)
Posts in 1924				0.513*** (0.138)
Observations	832	832	832	832
Bandwidth (km)	200.00	200.00	200.00	200.00
Clusters	18	18	18	18
<i>Panel B: RC Concession</i>				
RC	-0.030* (0.016)		-0.007 (0.006)	0.151* (0.077)
Posts in 1924				1.940*** (0.023)
Observations	154		154	154
Bandwidth (km)	100.00		100.00	100.00
Clusters	18		18	18

Notes. The table reports cross-sectional regression results for the number of facilities or missionary posts in each 20 km × 20 km grid cell. Panel A reports results for the UMHK concession (DRC) within a 200 km bandwidth. Panel B reports results for the RC concession (Zambia) within a 100 km bandwidth. Column (4) controls for the number of missionary posts in 1924. Standard errors are clustered at the territory level and reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Monetisation of the Family. Under the UMHK’s Labour Stabilisation Programs, the family unit was explicitly monetised. The corporation provided bride prices, payments for newborns, and family-sized housing to secure a long-term workforce. This transformation was a documented reality: the female-to-male ratio in UMHK compounds rose from 28% to over 80% (Juif and Frankema, 2018). By making access to substantial non-wage benefits conditional on the presence of a spouse, the UMHK framework effectively elevated the wife from a dependent to a structural ‘shareholder’ in the labour contract. I argue that this integration of women into the formal economic architecture established an initial endowment of female bargaining power. In a collective household framework, this arguably shifted the

female threat point: the cost to the husband of household dissolution was no longer just the loss of a partner, but the loss of the paternalistic premium itself. By explicitly financing the domestic unit, the UMHK inadvertently lowered the economic threshold for female autonomy, creating a structural precedent for female financial decision-making which proved durable.

Circular Migration. This stands in sharp contrast to the Rhokana Corporation’s model. The RC’s circular migration strategy actively discouraged permanent settlement to prevent “detrribalisation,” fostering a demographic imbalance heavily skewed toward male labourers. Historical and anthropological literature (e.g. [Chauncey, 1981](#); [Parpart, 1986](#); [Ferguson, 1999](#)) highlights how these temporary, male-dominated compounds created a high demand for commercial sex work and informal domestic services, leading to precarious gender relations. Consequently, the RC model lacked the institutional mechanisms that elevated female status in Katanga. This historical shock to the household equilibrium appears to have lowered the cost of female autonomy over the long run, manifesting today in the significantly higher propensity for women in the former UMHK concession to control household purchasing decisions and access modern contraception.

5.4 Urbanisation and Agglomeration

The final and perhaps most durable legacy of the UMHK’s settlement policy was the creation of dense urban clusters. Urbanisation can generate significant welfare gains through agglomeration economies ([Duranton and Puga, 2004](#); [Melo et al., 2009](#); [Donovan et al., 2024](#)), arising from shared infrastructure, labour pooling, and knowledge spillovers ([Puga, 2010](#)). Unlike the state, which lacked the fiscal capacity to subsidise urban density, the UMHK internalised these fixed costs (housing, sanitation) to solve the coordination failure of settlement.

Panel B of Table 6 reveals that the former UMHK concession area is persistently more urbanised relative to its surroundings today than the former RC concession area. This suggests that superior development outcomes may be explained by a greater capacity to realise

agglomeration benefits. Panel B provides further evidence, indicating that modern utility infrastructure in the UMHK area is more widespread. UMHK’s investments in healthcare and housing embedded an urban character that has persisted well beyond the colonial period.

Furthermore, the urban character of the former RC concession proved fragile. [Mususa \(2012\)](#) documents a process of “villagisation” in Zambian mining towns following the industry’s decline, where the withdrawal of corporate welfare forced residents to revert to rural subsistence strategies like charcoal burning and well-digging to survive. Unlike the resilient urban structure in the Congo, the Zambian urban fabric deteriorated significantly once the golden era of mine-provided services ended.

To formally test urbanisation as the primary mediating channel, [Table 8](#) repeats the baseline analysis while controlling for a contemporary ‘Urban Household’ indicator. The results are striking: after controlling for current urban status, the UMHK concession coefficient becomes small and statistically insignificant for virtually all major development outcomes (except wealth score). These results imply that the historical, corporate-driven divergence in settlement patterns and urbanisation largely account for the positive development discontinuities observed today. This indicates that urbanisation acts here not merely as a spatial container for current infrastructure, but as a path-dependent vessel for the accumulation of human capital. The UMHK did not simply build towns; it curated a specific demographic density that lowered the cost of service delivery and sustained the demand for education long after the corporation’s departure.

Congestion. The benefits of agglomeration are not unambiguously positive. Rapid urbanisation often generates environmental and social externalities—such as congestion and disease transmission—which can offset productivity gains ([Grover et al., 2023](#)). This concept of offsetting dis-economies may explain the ambiguous health findings in [Table 6](#), where the discontinuity for children’s height was less pronounced than for wealth. This is consistent with a model where the UMHK’s urbanisation drive simultaneously generated positive

Table 8: UMHK Concession Urban Development Outcomes

	Baseline (1)	Mechanism (2)	Baseline (3)	Mechanism (4)
<i>Panel A: Asset Wealth</i>				
	Wealth index		Log(wealth score)	
UMHK	2.054*** (0.789)	0.312 (0.210)	17.099*** (6.189)	1.876*** (0.665)
Urban Household		2.267*** (0.168)		19.982*** (0.818)
Observations	8,992	8,522	8,992	8,206
Clusters	55	55	54	54
Bandwidth (km)	121.59	106.20	122.33	98.97
<i>Panel B: Education</i>				
	Years of education		Literacy	
UMHK	2.305** (0.913)	0.252 (0.370)	0.337** (0.140)	0.015 (0.016)
Urban Household		2.576*** (0.325)		0.409*** (0.047)
Observations	8,962	8,962	2,372	2,458
Clusters	55	54	55	54
Bandwidth (km)	118.84	123.62	111.49	116.56
<i>Panel C: Health</i>				
	Woman ht./age pctl.		Child ht./age pctl.	
UMHK	15.769** (6.865)	2.127 (1.634)	17.687* (9.766)	12.747 (8.326)
Urban Household		17.197*** (3.593)		6.618 (6.077)
Observations	808	808	867	823
Clusters	52	52	53	53
Bandwidth (km)	110.29	115.39	117.71	114.16

Notes. The table reports regression discontinuity estimates testing urbanization as a mediator. Columns (1) and (3) reproduce the baseline optimal bandwidth results. Columns (2) and (4) include an indicator for urban households to test the mechanism. Standard errors are clustered at the DHS cluster level and reported in parentheses. All regressions use a local linear specification with a triangular kernel and control for sex, age, age squared, and survey year. *Urban Household* is a binary indicator for whether the DHS cluster is classified as urban. See previous tables for definitions of outcome variables.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

agglomeration economies (boosting wealth) and negative congestion externalities (dampening health gains), further explored in Appendix A.6. While the UMHK model successfully amassed (human) capital, its infrastructural legacy relied heavily on communal sanitation – such as shared taps and latrines – which, under conditions of modern high density, have transformed from corporate amenities into potential vectors for communicable disease, partially eroding the physical growth benefits associated with higher household wealth.

Taken together, these findings reveal that colonial corporate labour policies generated legacies primarily through their influence on spatial formation. While wage coordination constrained direct compensation, divergent colonial ideologies resulted in contrasting approaches to settlement. These institutional differences fostered a more urbanised, resilient community in the UMHK concession area. Furthermore, as shown in Appendix A.7, extractive “mine-to-port” transportation infrastructure does not explain this persistent prosperity, reinforcing the centrality of the labour-urbanisation channel. This analysis underscores the importance of accounting for the interactions between labour policy, gender relations, and urbanisation in both historical and contemporary development contexts.

6 Conclusion

This paper disentangles the long-run developmental effects of mineral wealth from the institutional contracts governing its extraction. The results demonstrate that the Union Minière du Haut-Katanga’s (UMHK) paternalistic “Labour Stabilisation” model generated persistent gains in human capital and wealth that endure decades after the firm’s nationalisation. In contrast, the Rhokana Corporation’s (RC) reliance on circular migration and a rigid colour bar left no comparable developmental legacy.

These findings indicate a specific mechanism of institutional persistence: path-dependent urbanisation. The divergence between the two concessions suggests that corporate policy can solve the coordination failure of city formation. By internalizing the fixed costs of settle-

ment—providing family housing, healthcare, and marriage bonuses—the UMHK effectively subsidised the density required for agglomeration economies to take root. This appears to have created a self-sustaining demand for public goods and education that outlasted the corporate incentives that birthed them, thereby converting temporary resource rents into permanent spatial advantages.

Furthermore, the results highlight the unintended social consequences of corporate design. By monetizing the family unit to secure a stable workforce, the UMHK unintentionally shifted the intra-household bargaining equilibrium. This initial endowment of female status—rooted in the firm’s need for reproductive labour—has persisted as a cultural norm, manifested today in greater female reproductive autonomy and reduced tolerance for domestic violence. This implies that corporate institutions may reshape social norms as profoundly as state interventions.

Finally, these historical lessons offer a critical perspective on modern extractive strategies. Contemporary enclave or “fly-in, fly-out” mining models, which prioritise operational efficiency and minimise local integration, structurally resemble the Rhokana Corporation’s approach (See [Haglund, 2009](#)). While such models maximise short-term returns to capital, this analysis suggests they fail to generate the agglomeration externalities necessary for sustained local development. The contrast between the UMHK and RC demonstrates that sustainable resource-led growth requires institutions that align corporate profit maximisation with long-term community stability. Ultimately, this paper provides micro-founded evidence that under specific constraints, private firms can mimic the state, delivering public goods and institutional quality that determine the long-run trajectory of development.

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Appendix

A.1 Timelines

Figures A.1 and A.2 provide detailed operational timelines that visually map the major historical milestones for the two concessionary companies: the Union Minière du Haut-Katanga (UMHK) and the Rhokana Corporation (RC). The first timeline traces the history of the UMHK from its establishment in 1906, through the introduction of its Labour Stabilisation Programs following the Great Depression, to its nationalisation in 1966. In parallel, the second timeline charts the history of the RC, which was formed through a merger in 1931 and operated until 1969.

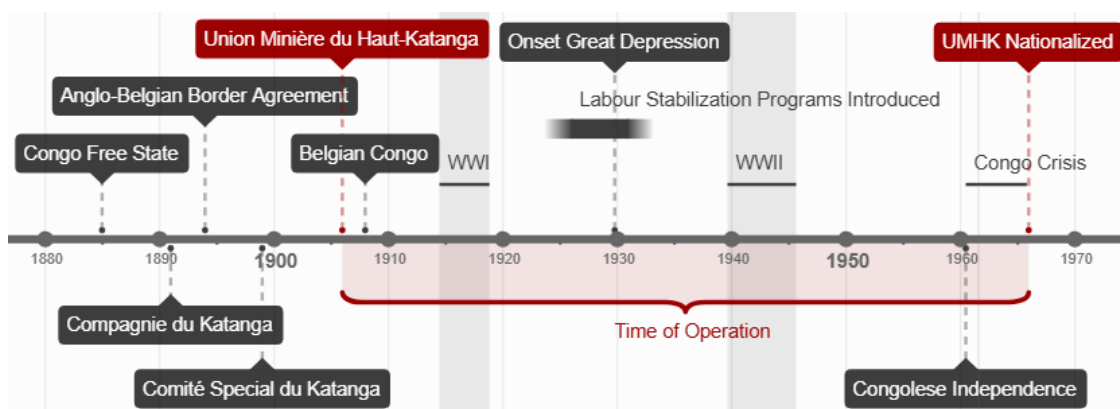


Figure A.1: UMHK Concession Timeline

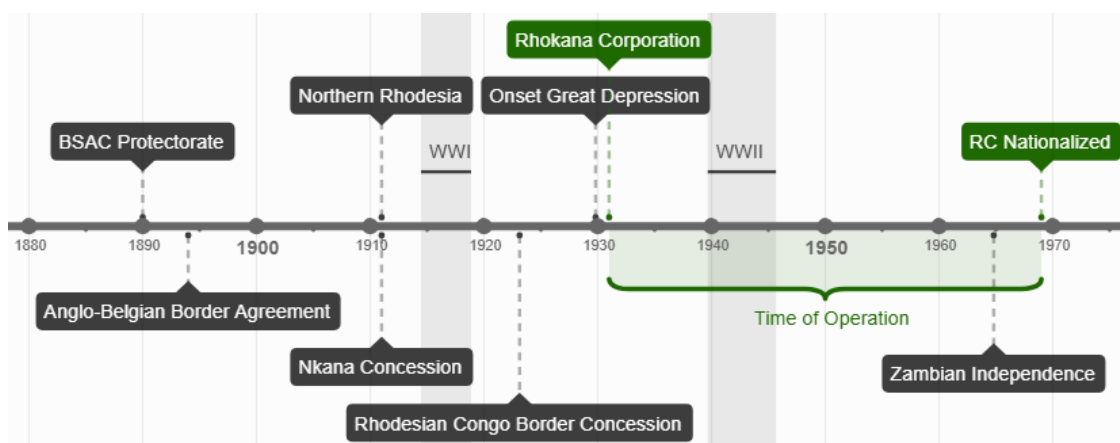


Figure A.2: RC Concession Timeline

A.2 Balance Tests

Figure A.4 shows the UMHK concession boundary alongside DHS clusters located within 200 km in the DRC; Figure A.5 shows the Nkana concession boundary and clusters within 100 km in Zambia.

Table A.2 presents summary statistics and estimates of geographic variables near the RC concession boundary. Further, for the RC concession boundary, Figure A.3 shows that it is trivial to prove the balance of pre-colonial characteristics. The concession (including the 100km bandwidth around it) lies wholly inside the Lamba ethnic group of the EA. Neither does the concession coincide with any contemporary administrative borders.

The central continuity assumption should also hold between the concessions of both colonies for the spatial difference-in-discontinuity design to be valid. This implies that there should not be any geographic or pre-colonial demographic difference-in-discontinuities for both the concession boundaries. Table A.2 presents difference-in-discontinuity balance tests for the geographic characteristics at 50km and 100km bandwidths, constructed analogous to equation 2. There is little sign of any large differences in geographic discontinuities around the concession boundaries. Only rainfall appears to be significantly different, but the magnitude of the difference makes it unlikely to threaten the diff-in-disc identification strategy.

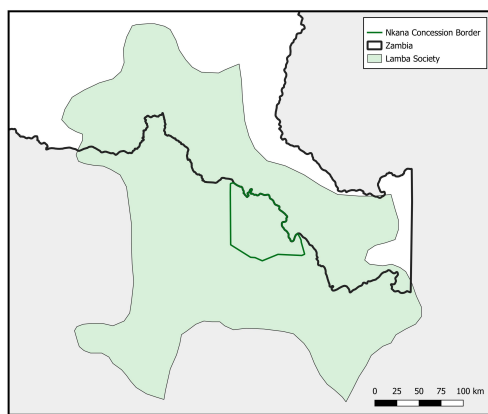


Figure A.3: RC Concession Boundary and the Lamba Ethnic Group

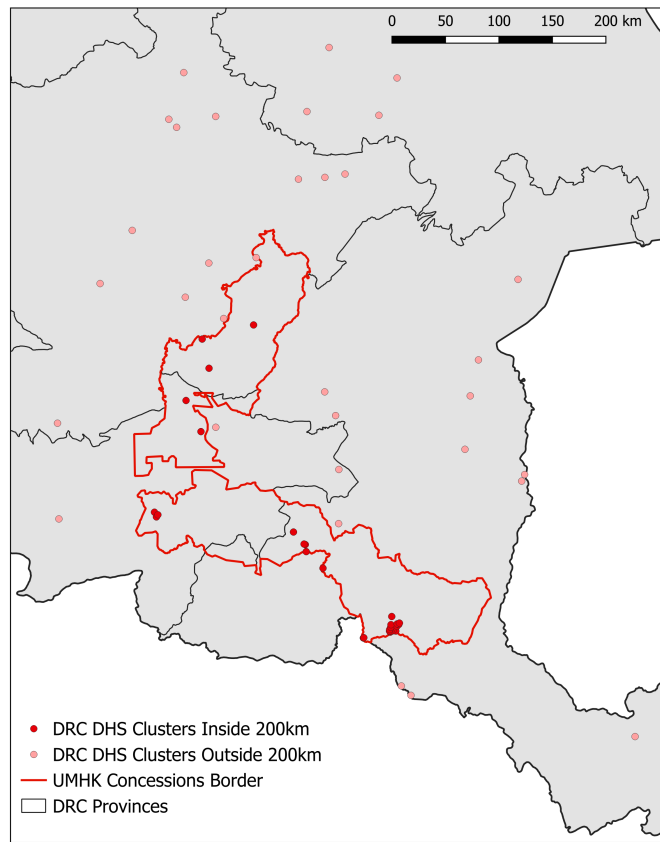


Figure A.4: UMHK Concession Boundary and DHS Clusters in the DRC

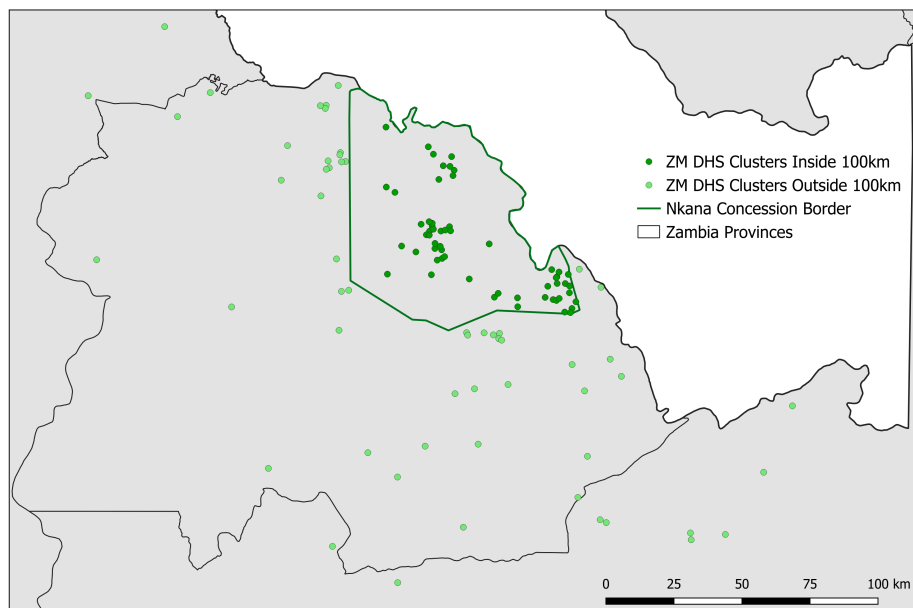


Figure A.5: Nkana Concession Boundary and DHS Clusters in Zambia

Table A.1: RC Concession Summary Statistics

Variable	Outside Concession			Inside Concession		
	Obs.	Mean	Std. Dev.	Obs.	Mean	Std. Dev.
<i>Panel A: Development Outcomes</i>						
Wealth index	7,233	3.071	1.346	6,970	4.238	0.921
Log(wealth score)	7,233	-2.482	10.820	6,970	7.605	8.463
Years of education	7,201	4.155	4.162	6,933	5.850	4.710
Literacy	2,622	0.752	0.432	2,890	0.891	0.312
Woman ht./age pctl.	1,377	27.207	25.006	1,556	29.813	25.793
Child ht./age pctl.	1,155	25.742	34.179	923	26.798	32.830
<i>Panel B: Women's Empowerment</i>						
Modern use (contra.)	470	0.560	0.497	488	0.619	0.486
Modern knowledge (contra.)	1,365	0.982	0.131	1,530	0.989	0.105
Arguing (dom. violence)	1,330	0.575	0.547	1,473	0.394	0.536
Refusing sex (dom. violence)	1,280	0.473	0.553	1,411	0.325	0.520
Remuneration	674	2.206	1.258	724	2.448	1.135
Household purchases	843	0.744	0.693	782	0.870	0.649
<i>Panel C: Urbanization</i>						
Nightlights	7,233	2.231	4.114	6,970	7.638	6.496
Built-up surface	7,233	0.176	0.300	6,970	0.401	0.280
Electricity	7,193	0.230	0.421	6,970	0.549	0.498
Water	1,363	0.278	0.448	1,530	0.622	0.485

Notes. The table reports summary statistics for the sample restricted to observations within 100km of the RC concession boundary. *Outside Concession* refers to units with distance < 0 (control group), and *Inside Concession* refers to units with distance > 0 (treatment group). See previous tables for detailed variable definitions.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.2: RC Concession Balance Test and Difference-in-Discontinuities

Variable	Panel A: RC Concession Balance (Bandwidth 50 km)			Panel B: DiDisc Estimates (BW 50 km) (BW 100 km)	
	Inside	Outside	RD Coeff.	DiDisc Coeff.	DiDisc Coeff.
	Mean (1)	Mean (2)	(SE) (3)	(SE) (4)	(SE) (5)
Elevation	1,266.40	1,260.77	−33.29 (29.16)	−18.16 (52.27)	−11.70 (41.95)
Ruggedness	72.54	75.40	0.57 (5.37)	−5.89 (13.59)	−3.56 (10.91)
Rainfall (avg., mm)	1,224.61	1,167.70	7.27 (7.74)	−23.87* (12.41)	−31.82*** (10.22)
Rainfall (st. dev.)	5.56	8.37	−1.42 (2.45)	0.92 (2.26)	2.79 (2.04)
Land suitability	96.91	96.99	−0.05 (0.04)	0.06 (0.11)	0.13 (0.10)
Malaria suitability	4.00	4.27	0.07 (0.44)	0.18 (0.67)	0.30 (0.53)

Notes. The unit of observation is a 20 km × 20 km grid cell. Panel A reports balance tests for the RC concession boundary. Columns (1) and (2) present means inside and outside the concession within a 50 km bandwidth. Column (3) reports the RD coefficient and standard error (in parentheses) using the optimal bandwidth procedure from [Cattaneo et al. \(2019\)](#). Standard errors are clustered at the territory level. Panel B reports Difference-in-Discontinuities (DiDisc) estimates. Columns (4) and (5) interpret the effect on the discontinuity of being at the UMHK border relative to the RC border, using bandwidths of 50 km and 100 km respectively.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.3 Concession Contamination

Figure A.6 provides additional historical context, showing the obsolete Rhodesia Congo Border Concession (RCBC), which fully contained the smaller Nkana concession. The paper's analysis focuses exclusively on the Nkana boundary, representing the Rhokana Corporation's effective sphere of influence. The larger RCBC boundary does not contaminate any findings, as virtually all data clusters used in the Nkana analysis lie entirely within its territory, precluding any confounding effects from the outer border.

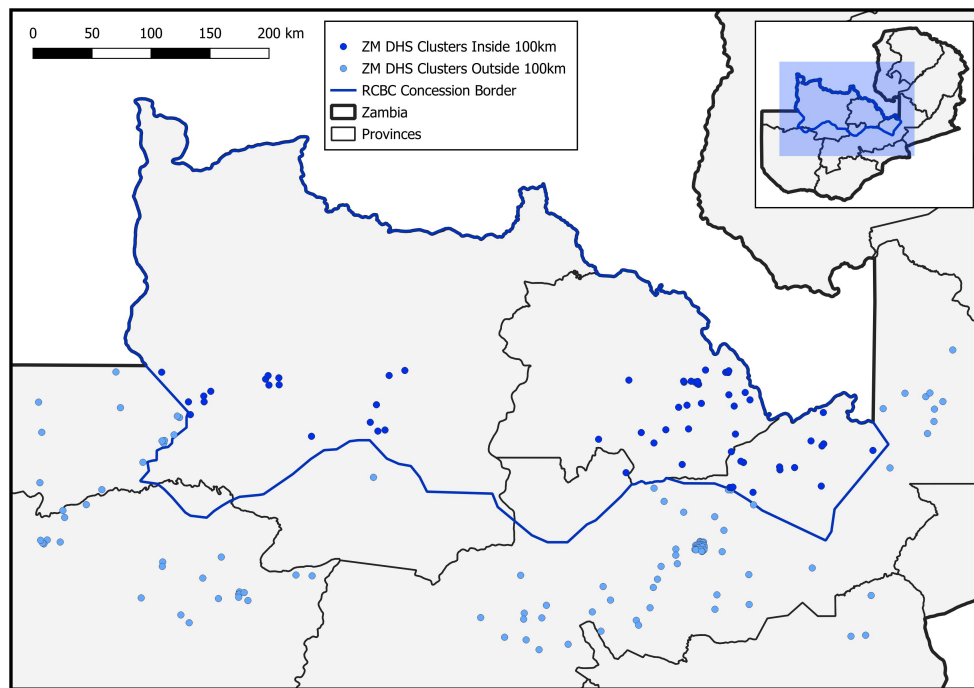


Figure A.6: RCBC Concession Boundary and DHS Clusters

A.4 Full Individual RD Regressions

This appendix section provides the complete regression discontinuity (RD) estimates for outcomes at the individual concession level. The following tables (A.3 and A.4) detail the specific discontinuities found at the borders of the UMHK and RC concessions for development, gender, and urbanisation related outcomes. These individual estimates are the component results that are synthesised and formally compared in the difference-in-discontinuities analyses in the main body of the paper.

A key finding of this paper, detailed in Table A.3, is the relatively consistent lack of statistically significant discontinuities at the Rhokana Corporation (RC) boundary (excepting domestic violence justifications for the wide bandwidth estimations only). This “null effect” across development and gender outcomes warrants careful interpretation. Given the RC’s documented extractive policies (its reliance on seasonal contracts, active discouragement of permanent family settlement, and enforcement of a rigid colour bar) one might have expected to find persistent negative developmental effects, similar to other extractive institutions in the literature.

However, the null finding is a comparative result. It signifies that the region inside the RC concession is not measurably different from the region just outside it. This implies that the baseline colonial framework in the rest of Northern Rhodesia was itself sufficiently non-developmental that the RC’s corporate model did not constitute a comparatively worse regime. This interpretation strengthens the paper’s central Diff-in-Disc finding: the observed divergence is not driven by a negative RC legacy, but by a positive and unique institutional legacy created by the UMHK’s Labour Stabilisation Programs.

Table A.3: RC Concession Development and Women's Outcomes

	Optimal Bandwidth (1)	Wide Bandwidth (2)	Optimal Bandwidth (3)	Wide Bandwidth (4)
<i>Panel A: Development Outcomes</i>				
	Wealth index		Log(wealth score)	
RC	-0.144 (0.350)	0.325 (0.285)	-0.212 (3.072)	3.300 (2.470)
Observations	11,228	14,203	11,464	14,203
	Years of education		Literacy	
RC	0.009 (0.529)	0.423 (0.471)	-0.039 (0.053)	0.041 (0.045)
Observations	11,416	14,134	4,498	5,512
	Woman ht./age pctl.		Child ht./age pctl.	
RC	1.210 (2.840)	1.169 (2.604)	3.873 (3.537)	4.058 (3.077)
Observations	2,678	2,933	1,658	2,078
<i>Panel B: Women's Outcomes</i>				
	Modern use (contra.)		Modern knowledge	
RC	0.094 (0.163)	0.130 (0.149)	0.002 (0.022)	0.002 (0.020)
Observations	1,849	2,039	1,683	2,039
	Domestic violence		Refusing sex	
RC	-0.035 (0.061)	-0.127** (0.050)	-0.066 (0.067)	-0.134** (0.053)
Observations	1,432	1,852	1,560	1,898
	Remuneration		Daily purchases	
RC	-0.091 (0.178)	0.202 (0.160)	-0.082 (0.068)	-0.003 (0.057)
Observations	1,024	1,407	602	666

Notes. The table reports regression discontinuity estimates for the RC Concession. Standard errors are clustered at the DHS cluster level and reported in parentheses. All regressions use a local linear specification with a triangular kernel and control for sex, age, age squared, and survey year. Optimal bandwidths are chosen using the MSE-minimizing procedure suggested by Cattaneo et al. (2019). See previous tables for full variable definitions.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table A.4: UMHK & RC Concession Urbanisation Outcomes

	Optimal Bandwidth (1)	Wide Bandwidth (2)	Optimal Bandwidth (3)	Wide Bandwidth (4)
<i>Panel A: UMHK Concession</i>				
	Nightlights		Built-up Surface	
UMHK	9.428*** (2.363)	9.479*** (2.406)	0.662*** (0.144)	0.654*** (0.142)
Observations	9,660	10,738	9,772	10,738
Bandwidth (km)	152.09	200.00	156.72	200.00
	Electricity		Water	
UMHK	0.751*** (0.174)	0.741*** (0.167)	0.667*** (0.190)	0.656*** (0.203)
Observations	9,772	10,738	9,174	10,738
Bandwidth (km)	160.08	200.00	132.49	200.00
<i>Panel B: RC Concession</i>				
	Nightlights		Built-up Surface	
RC	0.430 (1.853)	1.867 (1.742)	-0.208* (0.119)	-0.070 (0.100)
Observations	11,090	14,203	11,464	14,203
Bandwidth (km)	42.14	100.00	45.44	100.00
	Electricity		Water	
RC	-0.074 (0.155)	0.055 (0.128)	-0.043 (0.135)	0.094 (0.118)
Observations	11,442	14,163	2,403	2,893
Bandwidth (km)	48.33	100.00	47.31	100.00

Notes. The table reports regression discontinuity estimates for agglomeration and infrastructure outcomes. Standard errors are clustered at the DHS cluster level and reported in parentheses. All regressions use a local linear specification with a triangular kernel and control for sex, age, age squared, and survey year. Optimal bandwidths are chosen using the MSE-minimizing procedure suggested by [Cattaneo et al. \(2019\)](#). *Nightlights* is the luminosity measured by VIIRS. *Built-up Surface* is an index (0–1) from the Global Human Settlement Layer. *Electricity* and *Water* are binary indicators for household access to electricity and piped water (including public taps), respectively.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.5 Alternative Specifications

The paper’s baseline findings demonstrate considerable robustness, holding firm against several alternative specifications of the regression discontinuity (RD) design. A primary check involves substituting the baseline local linear polynomial with a local quadratic polynomial to ensure the results are not driven by the specific functional form. Table [A.5](#) confirms that the positive and statistically significant discontinuities in wealth, education, and health at the former UMHK boundary persist under the quadratic model.

Similarly, Table [A.5](#) also shows that the null effects observed at the RC boundary also hold, with no significant discontinuities emerging. Furthermore, the results are stable across different bandwidth selection procedures. The main analysis consistently presents findings using both an optimal bandwidth calculated via a Mean Squared Error minimizing procedure and a wider, fixed bandwidth. The consistency of the results across these different bandwidths demonstrates that the findings are not an artifact of a specific sample size around the cut-off.

Table A.5: UMHK & RC Concession Quadratic Polynomial Specifications

	Optimal Bandwidth (1)	Wide Bandwidth (2)	Optimal Bandwidth (3)	Wide Bandwidth (4)
<i>Panel A: UMHK Concession</i>				
	Wealth Index		Log(Wealth Score)	
UMHK	1.624*** (0.578)	1.632*** (0.465)	15.146*** (4.921)	14.805*** (3.922)
Observations	8,522	10,738	8,522	10,738
	Years of Education		Literacy	
UMHK	1.946*** (0.603)	2.050*** (0.501)	0.265*** (0.097)	0.311*** (0.081)
Observations	8,493	10,704	2,324	3,025
	Woman Ht./Age Pctl.		Child Ht./Age Pctl.	
UMHK	11.561** (5.540)	13.482*** (4.348)	10.173 (6.230)	10.433** (5.112)
Observations	789	1,022	867	1,062
<i>Panel B: RC Concession</i>				
	Wealth Index		Log(Wealth Score)	
RC	-0.144 (0.350)	0.325 (0.285)	-0.212 (3.072)	3.300 (2.470)
Observations	11,228	14,203	11,464	14,203
	Years of Education		Literacy	
RC	0.009 (0.529)	0.423 (0.471)	-0.039 (0.053)	0.041 (0.045)
Observations	11,416	14,134	4,498	5,512
	Woman Ht./Age Pctl.		Child Ht./Age Pctl.	
RC	1.210 (2.840)	1.169 (2.604)	3.873 (3.537)	4.058 (3.077)
Observations	2,678	2,933	1,658	2,078

Notes. The table reports robustness checks using a *quadratic* polynomial specification. Standard errors are clustered at the DHS cluster level and reported in parentheses. All regressions use a triangular kernel and control for sex, age, age squared, and survey year. Optimal bandwidths are chosen using the MSE-minimizing procedure suggested by [Cattaneo et al. \(2019\)](#). Panel A reports results for the UMHK concession; Panel B reports results for the RC concession. See previous tables for full variable definitions.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.6 Urbanisation and Congestion

While the main analysis highlights the positive economic returns of agglomeration, rapid urbanisation can concurrently generate negative externalities – such as congestion and increased disease transmission – that may partially offset development gains. Table A.6 investigates this mechanism by examining infrastructure risk and the disease environment.

Infrastructure Risk. Panel A reveals that households within the former UMHK concession are significantly more likely to rely on high-risk communal infrastructure, specifically shared toilets and public taps, compared to those outside the boundary. These facilities serve as proxies for urban crowding and are vectors for disease.

Disease Environment. Consistent with higher density, Panel B presents positive (though insignificant) coefficients for communicable diseases like diarrhoea and coughing. In contrast, Panel C suggests that children experience less fever, and have an overall lower mortality rate inside of the former UMHK concession. The negative coefficient for fever is likely driven by the lower prevalence of malaria in dense urban environments.

This preliminary evidence of congestion costs, coupled with insignificant but large coefficients on childhood disease occurrence, provides a plausible explanation for the observed pattern where the positive discontinuity in health outcomes (such as child height) is smaller in magnitude than the discontinuity in wealth. It suggests that while the UMHK's settlement policies generated economic surplus through density, that same density introduced countervailing health risks that dampened physical development.

Table A.6: UMHK Concession Congestion Outcomes

	Optimal Bandwidth (1)	Wide Bandwidth (2)	Optimal Bandwidth (3)	Wide Bandwidth (4)
<i>Panel A: Infrastructure Risk</i>				
	Public Tap		Shared Toilet	
UMHK	0.282** (0.143)	0.288* (0.165)	0.124* (0.064)	0.099* (0.051)
Observations	135,458	135,458	135,487	135,487
Clusters	54	61	54	61
Bandwidth (km)	130.78	200.00	133.01	200.00
<i>Panel B: Communicable Disease</i>				
	Diarrhea		Cough	
UMHK	0.035 (0.029)	0.023 (0.018)	0.133 (0.082)	0.106 (0.079)
Observations	23,306	23,306	23,294	23,294
Clusters	54	61	54	61
Bandwidth (km)	118.52	200.00	116.78	200.00
<i>Panel C: Mortality</i>				
	Child Survival		Fever	
UMHK	0.052 (0.033)	0.069** (0.033)	-0.076 (0.073)	-0.100 (0.072)
Observations	25,978	25,978	23,299	23,299
Clusters	53	61	55	61
Bandwidth (km)	100.30	200.00	118.90	200.00

Notes. The table reports regression discontinuity estimates testing congestion and disease mechanisms. Standard errors are clustered at the DHS cluster level and reported in parentheses. All regressions use a local linear specification with a triangular kernel and control for sex, age, age squared, and survey year. Optimal bandwidths are chosen using the MSE-minimising procedure suggested by [Cattaneo et al. \(2019\)](#). *Public Tap* is an indicator for households relying on a shared public tap/standpipe. *Shared Toilet* is an indicator for using an improved toilet shared with other households. *Diarrhea*, *Cough*, and *Fever* are indicators for symptoms experienced by the child in the last two weeks. *Child Survival* is an indicator equal to 1 if the child is alive.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

A.7 Transportation Infrastructure

One potential explanation for persistent differences in development is colonial-era transportation infrastructure. Railways and roads were often central to colonial economic strategies (Storeygard, 2016; Jedwab and Moradi, 2016; Donaldson, 2018), though their impacts on long-run development vary depending on local context.

Both the UMHK and RC concessions relied on rail infrastructure to export minerals, but neither region maintained strong domestic connectivity after independence. The Benguela Railway, a key route for UMHK mineral exports via Angola, was later destroyed during the Angolan Civil War (Figure A.7). Road infrastructure deteriorated significantly in the DRC – illustrated by the common notion that “a journey that took one hour in the colonial period now takes a full day” (Van Reybrouck, 2010).

Table A.7 compares changes in road networks over time. Column (1) shows that the former UMHK area experienced a sharper deterioration than its surroundings. In contrast, column (2) shows modest improvements in the RC area’s roads. Yet this is not reflected in current differences in development outcomes.

These patterns are consistent with previous findings that colonial “mine-to-port” infrastructure often failed to generate broad-based local development (Bonfatti and Poelhekke, 2017), and therefore may not account for the long-run divergence. Instead, they reinforce the interpretation that labour policies and urbanisation – rather than extractive transport routes – are the primary mechanisms of persistence in this setting.



Figure A.7: Map of the Benguela Railway

Table A.7: UMHK & RC Concessions Infrastructure Development

	UMHK Concession Roads 2010 (km) (1)	RC Concession Roads 2020 (km) (2)
Concession	-3.851** (1.687)	7.119* (4.268)
Road network in 1968	0.812*** (0.055)	1.122*** (0.102)
Observations	832	154
Bandwidth (km)	200.00	100.00

Notes. The table reports cross-sectional regression results for road network length (km) in each 20 km $\times$ 20 km grid cell. Results are only meant as descriptive falsification, showing that road networks did not improve more inside UMHK than outside, hence are unlikely to drive the main development gap. Column (1) analyses the UMHK region (DRC) measuring roads in 2010. Column (2) analyses the RC region (Zambia) measuring roads in 2020. Both models control for the historical road network in 1968. The concession coefficients capture the relative change in infrastructure over the post-colonial period. Standard errors are clustered at the territory level and reported in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$